Unveiling the Failure of Positive Selection

Dongkyu Chang[1] Duk Gyoo Kim[2] Wooyoung Lim[3]

[1]Department of Economics and Finance, CityUHK

[2]School of Economics, Yonsei

[3]Department of Economics, HKUST

August 28, 2026

KES Summer

Summary First

Question: Theory (Board and Pycia, 2014) says when a buyer (whose valuation is private) has an outside option, a seller can enjoy the largest profit. It requires some logical processes. *When and in what sense do people fail to do so?*

Method: We build a two-round version of the game and characterized the unique PBE (with introducing Nature's epsilon override) featuring positive selection. We then experimentally examine the (failure of) logical processes.

Findings: Very systematic failure of equilibrium reasoning, from the earliest stage. A substantial fraction (42%) of the round-1 price offers, which shouldn't be rejected, are rejected. Only a small (if not zero) proportion of the sellers updated their belief in line with the idea of positive selection. Four alternative explanations are ruled out by additional experimental evidence.

Experiment

Let's jump to it due to time constraints

Consider two-person (at most two-round) bargaining.

- A buyer has a private value $v \in \{v_L = 70, v_M = 240, v_H = 500\}$, equally likely.
- A seller makes price offer $p_t \in \{10, 180, 440\}$ in round t, then a buyer accepts it (A), takes an outside option $w = 50$ (O), or rejects both price offer and the outside option (R).
- When A or O, Nature overrides with probability $\epsilon = 0.001$, replacing it with R.
- In case of R or Nature's override, the game moves on to the next bargaining round with prob. 0.8. The game is terminated with prob 0.2.
- When the game ends with A in round t, a buyer earns $v - p_t$, and a seller earns p_t.
- When the game ends with O, a buyer earns $w = 50$, and a seller earns 0.
- When the game ends with R or is terminated, both earn 0.

Belief Reporting

Whenever the subjects move to the second round, the sellers were asked to report their beliefs.

<u>Your Task as a Seller in Round 2</u>: Before submitting a new price offer, report how you believe the buyer's value, by filling out the following sentence.

I believe that the value of the buyer paired in this match is
70 with a (____)% of chance,
240 with a (____)% of chance,
500 with a (____)% of chance.

The three numbers must sum up to 100. **The reported probabilities will appear in your decision screen but will not be shared with the buyer.**

Full Commitment Benchmark

With full commitment power, the seller commits to

$$p_1 = p_2 = p_w^* = 440 := \arg \max_{p \in \{70,180,440\}} \sum_{v: v-w \geq p} p \cdot q(v)$$

(Among (1) $p_w = 70$ to sell the item with prob. 1, (2) $p_w = 180$ to sell it with prob. $2/3$, and (3) $p_w = 440$ to sell it with prob $1/3$, 440 yields the largest profit.)

- The high-type buyer accepts $p_w^* = 440$ in round 1.
- The other buyer types exercise the outside option immediately.
- No inter-temporal pricing and no delay.

Theoretical Predictions

Proposition

In a unique Perfect Bayesian equilibrium,

(i) *The seller's equilibrium offer is* $p_w^* = 440$.

(ii) *The buyer accepts the seller's offer* p *in any round if and only if* $v - p \geq w$*; otherwise, exercises the outside option immediately.*

(iii) *No delay occurs with probability* $1 - \epsilon$ *in the equilibrium both on and off the equilibrium path.*

(iv) *If ever moved to round 2, the seller's posterior belief* $\hat{q}(v|p_1)$ *is identical to the prior* $q(v)$*, so* $p_2 = p_w^* = 440$.

Proof: Hold on. Wait for Hypotheses.

Hypotheses (1/5): The First-Order Positive Selection

Each step of the proof of Proposition will be associated with a testable hypothesis.

- The "minimal" rationality: The buyer with $v_L = 70$ should never delay. The lowest possible price offer $p_1 = 10$ should be accepted, and for other price offers, taking the outside option and securing 50 is strictly better.
- If the game moves on to round 2, then the seller must believe that the low type remains only because of ϵ. This leads to

Hypothesis (First-order positive selection)

No low-type buyers choose to delay. If ever moved to the second round, the posterior belief of high/middle-type buyer is weakly greater than the posterior belief of low-type buyer, for any price offer in round 1. $\hat{q}(v_L|p_1) \leq \min\{\hat{q}(v_M|p_1), \hat{q}(v_H|p_1)\}$.

Hypotheses (2/5): The Second-Order Positive Selection

- Given the first-order positive selection, a rational seller will never offer $p_L = 10$ in round 2. (For simplicity, think $\epsilon = 0$. If you know only middle- and high-type buyers are remaining, there is no point of offering 10 in round 2.) The lowest possible offer p_2 is $p_M = 180$.
- Then, by following the same reasoning for no delay of the low type, the middle type also finds it strictly suboptimal to delay the negotiation to round 2.

Hypothesis (Second-order positive selection)

No middle-type buyers choose to delay. If ever moved to the second round, the posterior belief of high-type buyer is weakly greater than the posterior belief of low/middle-type buyer, for any price offer in round 1. $\hat{q}(v_L|p_1) = \hat{q}(v_M|p_1) \leq \hat{q}(v_H|p_1)$.

Hypothesis (3/5): The Third-Order Positive Selection

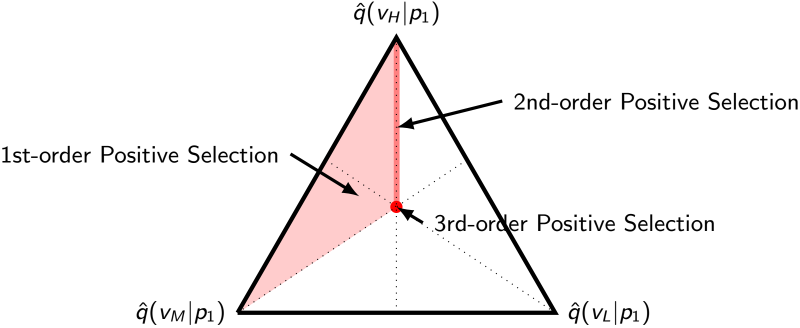
Recall that the full commitment price is $p^* = p_H = 440$.

- Given the first- and second-order positive selections, the posterior belief $\hat{q}$ weakly FOSD the prior belief q.
- The seller's 2nd(final)-round optimal offer must be greater than p^*, but since we have only three price alternatives, $p_2 = p_H$, implying that the high type has no reason to delay.

Hypothesis (Third-order positive selection)

In case $p^* = p_H$, no high-type buyers choose to delay. If ever moved to the second round, the posterior belief is the same as prior, $\hat{q}(v_L|p_1) = \hat{q}(v_M|p_1) = \hat{q}(v_H|p_1)$.

Summarizing the first three hypotheses



Positive Selections and Posterior Beliefs

- The earlier hypothesis *nests* the later one.
- If our experimental data do not support the theoretical predictions, these three hypotheses can provide us clear identification *from where* subjects fail.

Hypotheses (4/5)

This regards the seller's rationality.

- Given any posterior beliefs $\hat{q}(\cdot)$ the seller reported, the seller faces a static profit maximization problem.

$$\max_{p_2 \in \{70,180,440\}} \sum_{v:v-w \geq p_1} p_2 \cdot \hat{q}(v|p_1) \tag{1}$$

- We expect that the seller is at least best responding to her own (perhaps incorrect) belief.

Hypothesis

Given the seller's reported posterior belief $\hat{q}(\cdot)$, the price offered in the second round maximizes the seller's expected payoff.

Hypotheses (5/5)

This regards the buyer's rationality.

- H1–H3 state that no buyers would choose to delay. This is the case when the buyer expects:

$$E[\delta \max\{v - p_2, w\}] \leq \max\{w, v - p_1\} \quad \forall v, p_1,$$

- which means that some buyers may reject p_1, based on her subjective (perhaps incorrect) belief.
- We have only three types and three price alternatives, so we check in which case the above inequality *can* be violated.

Hypotheses (5/5)

$E[\delta \max\{v - p_2, w\}] \leq \max\{w, v - p_1\}$?

$v \setminus p_1$	p_L	p_M	p_H
v_L	always hold	always hold	always hold
v_M	always hold	can be violated	can be violated
v_H	always hold	can be violated	can be violated

Validity of not expecting "too low price" in Round 2

In words, if our experimental data shows some "rejections," then it is most likely in the v–p_1 pair shaded in red, somewhat likely in the pair shaded in blue, and not likely in other pairs. This leads

Hypothesis

Buyers with v_M or v_H are more likely to reject p_H, somewhat likely to reject p_M, and not likely to reject p_L. Buyers with v_L are not likely to reject any price offer in round 1.

Experiment: Basic Procedure

- oTree + Zoom real-time online experiment
- Turning on their video was a strict requirement
- HKUST, English
- 5 sessions each for *M90*, *M240*, and *M420*.
- $106 + 100 + 88 = 294$ participants
- Ten matches
- Random matching, between-subject design
- On average, HKD 115 ($\approx$ USD 16) including HKD 40 show-up payment
- Online bank transfer via the autopay system of HKUST

Results: Overview

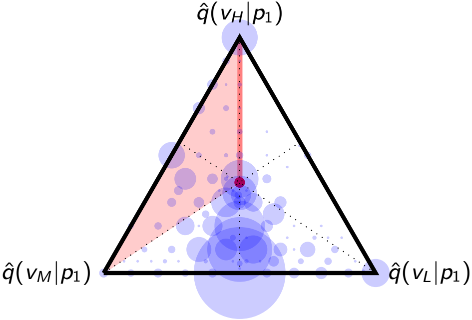
	$M240$
Avg.Offer (Theory)	295.86 (440)
%Reject_v_L (Theory)	35 (0)
%Reject_v_M (Theory)	60 (0)
%Reject_v_H (Theory)	50 (0)
Avg.SellerPayoffs (Theory)	87.94 (146.67)
Avg.BuyerPayoffs (Theory)	119.20 (53.33)

Table 1: Summary of Experimental Findings

Substantial differences:

- Rejections are pervasive.
- Avg.Offer is much smaller than what theory predicts.
- Avg.SellerPayoffs is much smaller than the equilibrium payoff.

Results 1–2

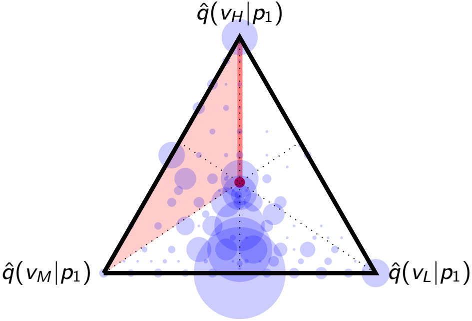


Observed posterior beliefs

36.64% of the posterior beliefs are rationalized in the first-order positive selection.

24.59% of the posterior beliefs (151 out of 614) are rationalized in the second-order positive selection.

Result 3



Observed posterior beliefs

Only 7.49% of the posterior beliefs (46 out of 614) are rationalized in the third-order positive selection.

Caveat: The third-order positive selection leads the posterior belief to be identical to the prior belief. They might be completely naïve.

Result 4

- 66.94% (411 out of 614) of p_2s were optimal from their subjective beliefs.
- Among sellers satisfying the 1st-order positive selection, 70.22% of p_2s were optimal. Among sellers satisfying the 2nd-order positive selection, 76.82% of p_2s were optimal.
- Among sellers satisfying the 3rd-order positive selection, less than half of p_2s were optimal. More than half of them seem to be the most naïve ones who kept the prior.

Result

Majority (66.94%) of the second-round price offers were optimal in the sense that the offer maximizes the expected profit calculated with their subjective beliefs. Higher-order reasoning on positive selection is positively associated with pricing optimality.

Result 5

$v \setminus p_1$	p_L	p_M	p_H
v_L	9% (1/11)	36% (76/211)	36% (102/280)
v_M	0% (0/3)	57% (137/240)	67% (151/227)
v_H	0% (0/7)	31% (72/231)	82% (214/260)

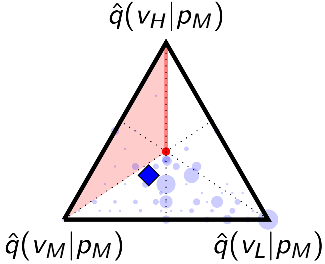
Proportions of Rejecting the First-Round Offer

Result

Some high- and middle-type buyers reject p_1 with expecting that p_2 would be more favorable to them. Some low-type buyers also reject p_1 where they could have been better off by exercising the outside option.

Further Result

Maybe Seller's posterior beliefs reflect Buyers' (perhaps irrational) rejection behaviors, and that may be the reason why they did not update the posterior belief in an equilibrium way? No.



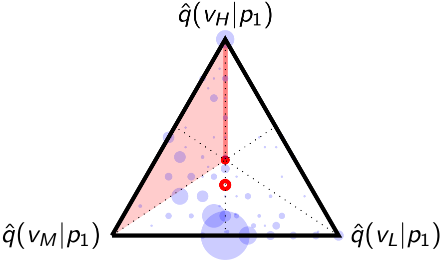
Observed and Empirical posterior beliefs, by p_1

The blue circles depict the reported posterior beliefs after the first-round price offers of p_M are rejected. A larger circle means more observations in the center of it. The diamond shape in each simplex is the posterior belief consistent with the empirically observed proportions of the rejections.

Supplementary Experiment 1

Is the failure due to a general inability at strategic reasoning? No

W/o an outside option, eqbm reasoning requires negative selection: High-type buyers are likely to accept p_1, with making the remaining demand pool consist of relatively lower types.



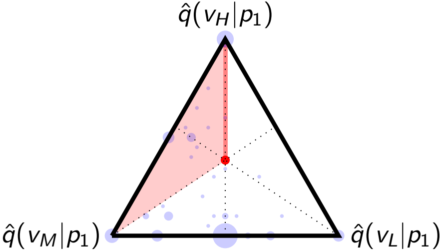
Observed posterior beliefs, without outside option

This figure shows that concentration (toward center-bottom) is consistent with negative selection and thus with equilibrium reasoning. Subjects are capable of the basic strategic reasoning underlying belief updating after a rejection.

Supplementary Experiment 2

Is the failure due to sellers' concern about buyer irrationality? No.

Although it is unlikely (see "Further Result" slide), the supplementary experiment addresses the concern more directly: p_L option is removed.



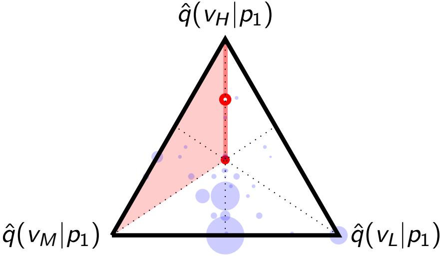
Observed posterior beliefs, without p_L

88% of L-type buyers took the outside option in round 1, recognizing that delaying is futile. Only 2% of H-type buyers did so, recognizing that the outside option is unattractive. Despite this, sellers' posterior beliefs after a r1 rejection continue to put substantial weight on the low type. This pattern provides direct evidence that the failure of positive selection is not driven by sellers' anticipation of buyer irrationality.

Supplementary Experiment 3

Is the failure due to other-regarding preferences? No.

We further validate this argument using the Amended Random Payment method. Under ARP, each participant's final payment is determined by an independently drawn random match, and participants are told *whether the current match determines their paired participant's payment*.



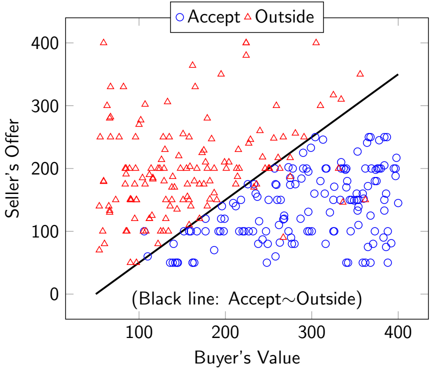
Observed posterior beliefs: ARP

If other-regarding preferences were driving our main results, rejection rates should fall and posterior beliefs should shift toward equilibrium under ARP. The same qualitative patterns—substantial rejections and posterior beliefs inconsistent with 1st-order positive selection—emerge in an environment where other-regarding motives are neutralized.

Supplementary Experiment 4

Is the failure due to sellers' limited understanding of optimal pricing? No

Additional 1-round bargaining (essentially ultimatum) experiment with a privately-informed buyer who holds an outside option.



The average offer is statistically indifferent from the commitment price. Most buyers accept the price offer rationally, without leaving a room for fairness concern.

Take-away Messages

- BP's prediction is robust in theory, in many ways.
- It builds upon many layers of rational belief updating, positive selection of the remaining demand pool.
- We found that a substantial fraction (41.77%) of p_1s are rejected, which shouldn't be observed in theory.
- Only about 7% of the sellers report the beliefs based on the 3rd-(or higher-)order positive selection.
- About half of them were naïve, meaning that few (if not none) thought in an "equilibrium" way.
- Failure of positive selection reasoning drives the inconsistency between theoretical predictions and observations.
- Our contribution is not only checking the validity of BP but also presenting and utilizing a way to decipher which level of positive selection reasoning fails.

Related Literature

Theory

- Board and Pycia (2014), Tirole (2016), Shimoji and Watson (1998), Coase (1972), Fudenberg et al. (1985), Gul et al. (1986), Hwang and Li (2017), Chang and lee (2022), Fanning (2026)

Experiment

- Kneeland (2015), Ho and Su (2013), Lim and Xiong (2021), Chang et al. (2026)